

PROBLEM STATEMENT

1: Hospital Doctor Shift Assignment

A hospital must prepare a duty schedule by assigning three doctors (D1, D2, and D3) to three different work shifts: Morning, Afternoon, and Night. The schedule should ensure that each shift is covered by exactly one doctor while satisfying all the given constraints related to doctor availability and shift preferences. The objective is to determine a valid and conflict-free assignment that fulfills every constraint without violating any scheduling rules.

2: Robot Grid Navigation

A mobile robot operates in a 5×5 grid environment and must travel from a designated Start (S) position to a Goal (G) position. The robot can move only in the four cardinal directions while avoiding blocked cells (obstacles). Since every movement has a fixed cost, the robot must identify the shortest and most efficient route that satisfies all movement constraints and safely reaches the destination.

3: Autonomous Rescue Robot

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The building is represented as a grid of interconnected rooms, where some paths are blocked due to debris and certain areas are considered risky. The robot has limited sensing capability and can observe only adjacent cells, making the environment partially observable. It must continuously update its knowledge while navigating, avoid obstacles, minimize the total travel cost, and safely reach the survivor's location. The challenge is to develop an intelligent decision-making strategy that enables the robot to perform efficient and safe rescue operations under the given environmental constraints.